

AHSAN SHAHZAD

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SUMMARY

AI/ML Engineer with 1+ year of hands-on experience building production GenAI systems, including RAG and multi-agent LangGraph workflows. Solid foundation in ML/DL, with practical experience in LLM evaluation, observability, and experiment tracking.

EDUCATION

B.S. in Software Engineering Comsats University Islamabad, Lahore Campus.	Sep 2021 – June 2025
Fsc Pre Engineering Government College University, Lahore.	Sep 2019 – June 2021

WORK EXPERIENCE

Associate Software Engineer / Folium AI Python FastAPI LangChain LangGraph Pinecone LLMs Docker Git LangSmith AWS	June 2025 – present
- Built an Agentic RAG chatbot within a Django backend for a crypto education platform, with a multi-format ingestion pipeline (preprocessing, chunking, Pinecone embeddings), conditional web search, and file handling; reached 95%+ accuracy on manual review and ~3 s time-to-first-token, monitored with LangSmith (tracing, p50/p95 latency, token cost). - Architected a multi-agent LangGraph system on FastAPI that turns messy medical billing documents into structured, validated data, with specialized agents for ingestion, validation, decision recommendations, letter generation, and vision-based extraction, coordinated through parallel pipelines, conditional routing, a cross-document consistency check, and Pydantic structured outputs. - Engineered its document processing layer on AWS Textract and S3 with async jobs, rebuilding tables and forms from raw OCR into LLM-ready markdown, and kept cost and failures in check by gating low-confidence scans, pairing concurrent LLM evaluation with rule-based checks, and isolating failures per document. - Developed a conversational assistant for a live youth savings platform: an intent router with multi-intent detection feeding a ReAct tool-calling agent over the Django backend, with location-aware and personalized search, multimodal input (images, documents, voice), guardrails, and model tiering (a lighter model for tool selection, a stronger one for responses). - Designed TOC and blog generation pipelines for an SEO content platform, combining SERP-driven competitor analysis, multi-pass rewriting, and web-search claim verification with automatic correction, using Claude/OpenAI model fallback. - Containerized services with Docker and set up GitHub Actions CI with pytest unit and regression tests (mocked LLM calls), coverage, ruff/mypy checks, pre-commit hooks, and container health checks.	

PROJECTS

Chest Cancer Classification – End-to-End DL project with MLOps Pipeline Python Keras DVC MLflow FastAPI Docker GitHub Actions AWS EC2	
- Built a config-driven, modular ML pipeline for CT-scan chest cancer classification, using transfer learning on VGG16 with data augmentation, early stopping, and learning-rate scheduling. Orchestrated it as a DVC pipeline (ingestion, base-model prep, training, evaluation) with parameter-aware stage caching, so only the stages affected by a config change re-run, giving fully reproducible experiments. - Tracked experiments with MLflow (parameters, metrics, versioned models in a model registry), served the registered model through a FastAPI inference API, containerized with Docker, and automated testing and deployment with a GitHub Actions CI/CD pipeline to AWS EC2.	
System Design Mentor – Hybrid RAG with Evaluation Suite LangGraph Pinecone OpenAI DeepEval LangSmith	
- Built a hybrid-search RAG assistant over five system design and ML books, combining dense and keyword (BM25) retrieval with a reranker and metadata-aware chunking. - Built an evaluation suite with golden datasets covering retrieval, answer quality, safety, latency, and cost, plus regression checks that compare each change against a baseline, and online scoring of live traces in LangSmith.	

SKILLS

Languages:	Python, TypeScript, JavaScript (ES6+), SQL, C/C++, Java
Frameworks & Libraries:	FastAPI, Django, Pydantic, React.js, Next.js, Node.js, Tailwind CSS
AI & ML Tools:	LangChain, LangGraph, LLM Evals, MCP , PyTorch, TensorFlow, Scikit-learn , Hugging Face,
Databases:	MongoDB, PostgreSQL, Supabase, Pinecone , chroma , Neo4j
Developer Tools:	AWS , Docker, Git, GitHub Actions (CI/CD), LangSmith, Jupyter Notebook, Jira , MLFlow, DVC
Core Skills:	RESTful APIs, Asynchronous Programming, OOP, DSA, Agile Development

CERTIFICATIONS

